

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 04

DATE : 21-11-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 04

Topics

DATE : 21 - 11 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : Circular Motion, W.P.E.

CHEMISTRY : GOC (Electronic Effect), Atomic Structure

BOTANY : Cell Division, Extension of Mendelian Genetics

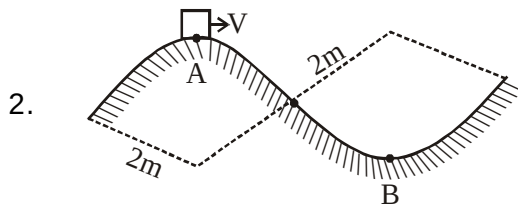
SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. Which of the following cannot be the safe speed of a car at a turn on a flat horizontal road of radius 120 m and friction coefficient 0.75.

(1) 72 km/hr (2) 108 km/hr
(3) 110 km/hr (4) 30 km/hr



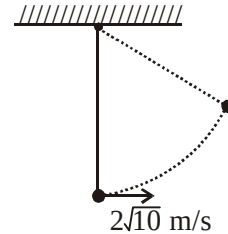
If a block of mass 100 kg passes from point A and B with same speed and it is found that difference between normal reactions at point B and point A is 400 N then find speed of car at point B.

(1) 2 m/s (2) 4 m/s
(3) 20 m/s (4) None of these

3. If a particles position vectors is $\vec{r} = 2\cos\theta\hat{i} + 2\sin\theta\hat{j}$. If particles speed at each instant is 4 m/s then find acceleration of particle.

(1) 4 (2) 18
(3) 16 (4) 8

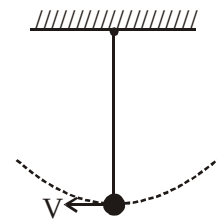
4. If a bob is given a velocity of $2\sqrt{10}$ m/s at lowest position then find tension in the string at extreme position. If mass of bob is 5 kg and string is 10m long.



(1) 20 N (2) 40 N
(3) 60 N (4) 10 N

5. If a pendulum string can withstand a maximum tension of 18 N at lowermost point. Find the maximum possible velocity (V) of bob of mass 1 kg at lowest position if length of string is 2m.

(1) 2 m/s
(2) 8 m/s
(3) 16 m/s
(4) 4 m/s

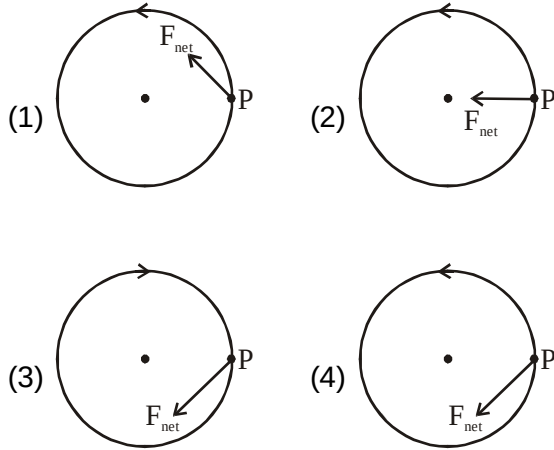


6. Find speed of a car on a banked road of angle 60° so that friction is not used if radius of track is $270\sqrt{3}$ m.

(1) 81 m/s (2) 90 m/s
(3) 30 m/s (4) None

SPACE FOR ROUGH WORK

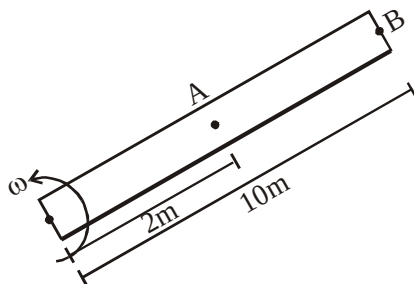
7. What can be the representation of net force on a particle (P) which is slowing down with a constant rate in circular motion.



8. If a particle at rest is subjected to angular acceleration $\alpha = (2t + 3t^2 + 2)$ (in rad/s^2) find its angular velocity (in rad/s) after 2 seconds.

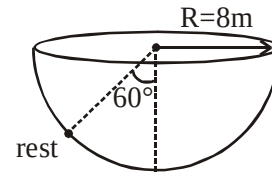
(1) 4 (2) 16 (3) 8 (4) 12

9. If a rod of length 10 m is rotating about a fixed point as shown in a horizontal plane with constant angular velocity of 5 rad/sec . then find ratio of linear velocities of point A to point B.



(1) 10 (2) 5 (3) 0.2 (4) 0.1

10. A ball is released from rest as shown in a frictionless hemispherical bowl. If mass of ball is 2 kg then find normal reaction when ball reaches lowest point.

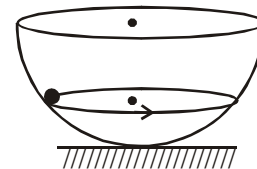


(1) 20N (2) 40N
(3) 60N (4) 80N

11. If frequency of revolution of a particle

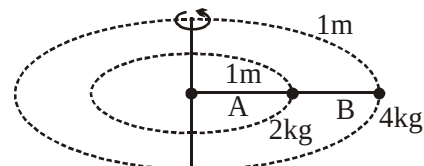
inside a frictionless bowl is $\left(\frac{5}{\pi}\right)$ R.P.S.

then find at what height from base particle is revolving if radius of bowl is 0.4 m.



(1) 0.1 (2) 0.3 (3) 0.2 (4) 0.35

12. If two particles are rotated in horizontal circle as shown if at an instant velocity of 4 kg is 4 m/s find ratio of tension in string A to the tension in string B.



(1) $\frac{4}{5}$ (2) $\frac{5}{4}$
(3) $\frac{1}{4}$ (4) 4

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13. For a particle starting from rest under constant angular acceleration of 2 rad/sec^2 . In a circle of radius 1 m . Find magnitude of net acceleration in m/s^2 after 2 sec .

(1) $\sqrt{260}$ (2) 4
(3) $\sqrt{20}$ (4) $\sqrt{256}$

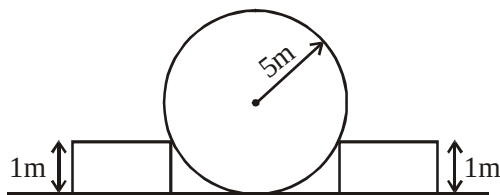
14. At what maximum distance from the centre a block of mass 2 kg must be placed on a turntable rotating with an angular speed of 4 rad/sec . So that it will not slide on surface. If friction coefficient is 0.8 .

(1) 0.5 m (2) 2 m
(3) 8 m (4) 4 m

15. A car moves safely on a frictionless banked road with a speed equal to the minimum safe speed in a death well of radius, which is half the radius of road and friction coefficient of 0.5 . Find banking angle of the road.

(1) 30° (2) 60°
(3) 45° (4) None of these

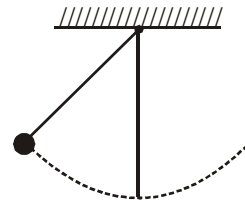
16. A sphere of radius 5 m is resting on two identical blocks of height 1 m as shown. Friction is absent every where. When sphere is about to touch the ground, its speed is V . At this instant speed of blocks :-



(1) $\frac{5V}{3}$ (2) $\frac{3V}{5}$

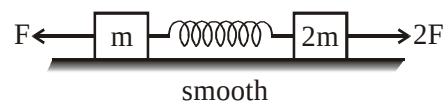
(3) $\frac{4V}{3}$ (4) $\frac{5V}{4}$

17. A pendulum bob is released from rest as shown then find incorrect statement.



- (1) work done by tension increases kinetic energy as it comes down.
(2) gravitational potential energy changes as bob moves.
(3) tension increases as bob moves in downward direction.
(4) velocity of bob is maximum at lowest position.

18. In the figure shown, find maximum elongation in spring, which initially is unstretched. Spring constant of spring is K :-

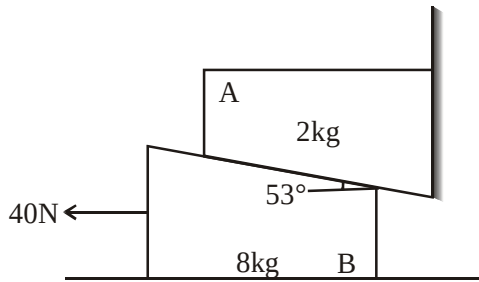


(1) $\frac{4F}{K}$ (2) $\frac{8F}{3K}$

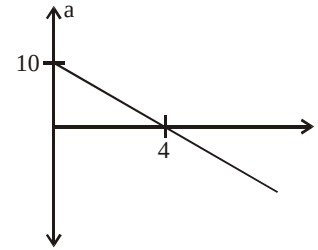
(3) $\frac{3F}{3K}$ (4) $\frac{F}{K}$

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19. In the figure shown, find acceleration of block A. (Neglect friction everywhere) :-



- (1) $\frac{7g}{13}$ (2) $\frac{9g}{13}$
(3) $\frac{9g}{26}$ (4) $\frac{10g}{13}$
20. A particle is projected at 60° to the horizontal with a kinetic energy K . The kinetic energy at the highest point is :-
(1) K (2) 0
(3) $\frac{K}{4}$ (4) $\frac{K}{2}$
21. A particle is projected with initial velocity of $(a\hat{i} + 2b\hat{j})$. Its range on horizontal plane is given by :-
(1) $\frac{ab}{2g}$ (2) $\frac{2ab}{g}$
(3) $\frac{4ab}{g}$ (4) $\frac{3ab}{g}$
22. The acceleration time graph of a particle moving along a straight line as shown in figure. At what time the final velocity is equal to initial velocity :-



- (1) 12 sec. (2) 4 sec.
(3) 8 sec. (4) 16 sec.
23. In the figure shown, find speed of x .
-
- (1) $2\sqrt{5}$ m/s
(2) $4\sqrt{2}$ m/s
(3) $2\sqrt{2}$ m/s
(4) 4 m/s
24. A weightless elastic rope of natural length 11 m is hanging from roof of height 12.5 m. A man of height 1.5 m and mass 55 kg is slowly descending from the top. What is distance between bottom of rope and ground when man's feet just touches the floor. It is given that a force of 10 N produces an elongation of 2 cm in the entire rope. Assume man's hand are holding the rope at the level of the head.
- (1) 200 mm (2) 300 mm
(3) 400 mm (4) 500 mm

SPACE FOR ROUGH WORK

25. A swimmer can swim in still water with 5 m/s. If a river is flowing with 4 m/s then find the time to cross the river of width 300 m if he has to reach to exactly opposite point :-

(1) 60 sec. (2) 75 sec.
(3) 100 sec. (4) $\frac{100}{3}$ sec.

26. A car moving at 160 km/h when crosses the marker-0, driver applies brake and reduces its speed uniformly to 40 km/h at marker-2. The markers are spaced at equal distances along the road as shown below.



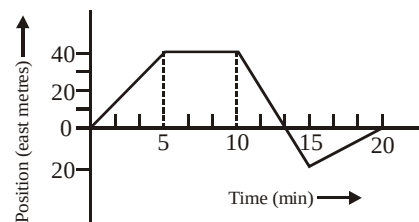
At which part of the track would the car have been traveling at instantaneous speed of 100 km/h ?

- (1) At marker-1
(2) Between marker-0 and marker-1
(3) Between marker-1 and marker-2
(4) More information required to decide
27. A bullet fired into a fixed target loses half of its velocity after penetrating 3 cm. How much further it will penetrate before coming to rest assuming that it faces constant resistance to motion :-
- (1) 1.5 cm
(2) 1.0 cm
(3) 3.0 cm
(4) 2.0 cm

28. A body A starts from rest with an acceleration a_1 . After 2 seconds, another body B starts from rest with an acceleration a_2 . If they travel equal distances in the 5th second, after the start of A, then the ratio $a_1 : a_2$ is equal to :-

(1) 5 : 9 (2) 5 : 7
(3) 9 : 5 (4) 9 : 7

29. A boy begins to walk eastward along a street in front of his house and the graph of his position from home is shown in the following figure. His average speed for the whole time interval is equal to :-



(1) 8 m/min (2) 6 m/min
(3) $\frac{8}{3}$ m/min (4) 2 m/min

30. The mass of the bob of a simple pendulum is m and it is raised to a maximum height h . The work done by gravity in moving it from one extreme end to the other is

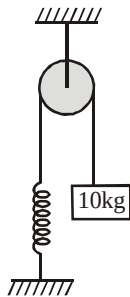
(1) $2 mgh$ (2) mgh
(3) $3 mgh$ (4) zero

31. Two spring A and B have spring constants K_A and K_B such that $K_B = 2 K_A$. The free ends of both the springs are stretched by same force. If energy stored in spring A is E then energy stored in spring B will be

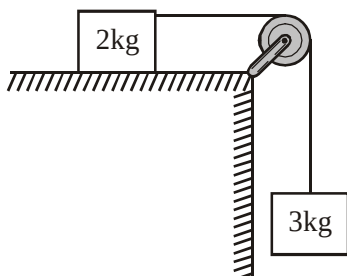
(1) $E/2$ (2) $2E$ (3) E (4) $4E$

SPACE FOR ROUGH WORK

32. For the system shown in figure the potential energy stored in the spring is ($K = 200 \text{ N/m}$)

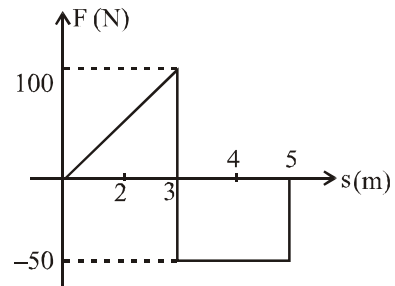


- (1) 10 J (2) 15 J (3) 25 J (4) 50 J
33. At a certain point, the potential and kinetic energies of a body of mass 4kg projected vertically up are 200 J and 600 J respectively. The maximum height reached by the body and the velocity with which it is projected from the ground are
(1) 20m, 20m/s (2) 20m, 10m/s
(3) 10m, 20m/s (4) 10m, 10m/s
34. A time varying power $P = 2t$ is applied on a particle of mass 3kg. The kinetic energy of the particle at $t = 4\text{ s}$ is
(1) 4 J (2) 8 J (3) 16 J (4) 24 K
35. If the blocks shown in the figure are released from rest, what is the power delivered by the tension in the string to the 2 kg block in 2 s ?



- (1) 72 watt (2) 144 watt
(3) 36 watt (4) 180 watt

36. An engine pumps water through a hose pipe. Water passes through the pipe and leaves it with velocity of 4m/s. The mass per unit length of water in the pipe is 10 kg/m. What is the power:-
(1) 640 watt (2) 320 watt
(3) 160 watt (4) 10 watt
37. In the given graph if mass of body $m = 10\text{ kg}$ and initial speed is 10m/s. Then find out KE of body after a displacement of 5m :-



- (1) 50 J (2) 400 J
(3) 550 J (4) 500 J
38. If K.E. of a body is decreased by 36%. Then % change in momentum is :-
(1) 20% ↓ (2) 20% ↑
(3) 40% ↓ (4) 40% ↑
39. If P.E. of two molecules is given as

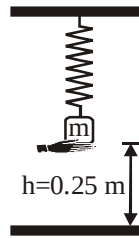
$$U = \frac{A}{r^{10}} - \frac{B}{r^5} \text{ where } A \text{ and } B \text{ are constant and}$$

'r' is separation between molecules then find equilibrium position in terms of r.

- (1) $\left(\frac{2A}{B}\right)^{-1/6}$ (2) $\left(\frac{A}{2B}\right)^{-1/5}$
(3) $\left(\frac{A}{B}\right)^{1/6}$ (4) $\left(\frac{2A}{B}\right)^{1/5}$

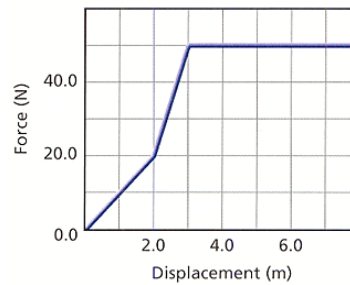
SPACE FOR ROUGH WORK

40. An ideal and massless spring ($k=10\text{N/cm}$) hangs from ceiling with a block which is held initially as shown in the figure. What should be mass of block (in kg) such that it just touches the floor when released from rest? Assume no tension in the spring when block is released.



- (1) 0.125 (2) 12.5
(3) 6.25 (4) 25
41. A fan keeps working for 30 minutes and uses energy of 90000 J, power consumed by fan is
(1) 3000 W
(2) 50 W
(3) 500 W
(4) 300 W
42. A boy walks up staircase of 20 m with a force of 600 N, time taken by boy is 8 minutes. power consumed by boy is
(1) 25 W
(2) 240 W
(3) 16 W
(4) 72.5 W

43.



Work Done by force from $x = 2\text{m}$ to $x = 4\text{m}$ is

- (1) 140 J (2) 70 J
(3) 35 J (4) 85 J
44. A particle moves on a rough horizontal ground with some initial velocity say v_0 . If $\frac{3}{4}$ of its kinetic energy is lost due to friction in time t_0 then coefficient of friction between the particle and the ground is :-
(1) $\frac{v_0}{2gt_0}$ (2) $\frac{v_0}{4gt_0}$
(3) $\frac{3v_0}{4gt_0}$ (4) $\frac{v_0}{gt_0}$
45. A variable force, given by the 2-dimensional vector $\vec{F} = (3x^2\hat{i} + 4\hat{j})$, acts on a particle. The force is in newton and x is in metre. What is the change in the kinetic energy of the particle as it moves from the point with coordinates (2,3) to (3,0) ? (The coordinates are in meters)
(1) -7 J (2) Zero
(3) +7J (4) +19 J

SPACE FOR ROUGH WORK

46. Select iso electronic set :-
 (a) Na^+ , H_3O^+ , NH_4^+
 (b) CO_3^{2-} , NO_3^- , HCO_3^-
 (c) P^{3-} , HCl , C_2H_6 , PH_3
 (d) N^{3-} , O^{2-} , F
 (1) a, b, d (2) b, c, d
 (3) a, b, c, d (4) a, b, c
47. The atom A, B, C have the configuration
 $\text{A} \rightarrow [\text{Z}(90) + n(146)]$, $\text{B} \rightarrow [\text{Z}(92) + n(146)]$,
 $\text{C} \rightarrow [\text{Z}(90) + n(148)]$ So that :-
 (a) A and C - Isotones
 (b) A and C - Isotopes
 (c) A and B - Isobars
 (d) B and C - Isobars
 (e) B and C - Isotopes
 The wrong statement's are:-
 (1) a, b only (2) c, d, e only
 (3) a, c, d only (4) a, c, e only
48. Which of the following statements concerning the two isotopes $_{13}\text{Al}^{27}$ and $_{14}\text{Si}^{27}$ are not incorrect :-
 (1) The aluminium isotope has more neutrons per atom than the silicon isotope
 (2) If $_{13}\text{Al}^{27}$ absorbs neutron, silicon isotope is formed
 (3) Both the isotopes have a atomic number 27
 (4) Both the isotopes contain 27 electrons per atom
49. Atomic weight of Ne is 20.2. Ne is mixture of Ne^{20} and Ne^{22} , Relative abundance of heavier isotope is :-
 (1) 90 (2) 20
 (3) 40 (4) 10
50. The uncertainty in position of an electron & helium atom are same. If the uncertainty in momentum for the electron is 32×10^5 , then the uncertainty in momentum of helium atom will be
 (1) 32×10^5 (2) 16×10^5
 (3) 8×10^5 (4) None
51. What is the ratio of the De-Broglie wave lengths for electrons accelerated through 200 volts and 50 volts :-
 (1) 1 : 2 (2) 2 : 1
 (3) 3 : 10 (4) 10 : 3
52. In an atom, for how many electrons, the quantum numbers will be , $n = 3$, $\ell = 2$,
 $m = +2$, $s = +\frac{1}{2}$:-
 (1) 18 (2) 6
 (3) 24 (4) 1
53. For the azimuthal quantum number (ℓ), the total number of magnetic quantum number is given by:-
 (1) $\ell = \frac{(m+1)}{2}$ (2) $\ell = \frac{(m-1)}{2}$
 (3) $\ell = \frac{(2m+1)}{2}$ (4) $\ell = \frac{(2m-1)}{2}$
54. Energy of atomic orbitals in a particular shell is in order:-
 (1) $s < p < d < f$
 (2) $s > p > d > f$
 (3) $p < d < f < s$
 (4) $f > d > s > p$

SPACE FOR ROUGH WORK

55. Spin angular momentum for electron :-

(1) $\sqrt{s(s+1)} \frac{h}{2\pi}$ (2) $\sqrt{2s(s+1)} \frac{h}{2\pi}$

(3) $\sqrt{s(s+2)} \frac{h}{2\pi}$ (4) None

56. An atom has 2 electrons in K-shell, 8 electrons in L-shell & 8 electrons in M-shell. The number of s-electrons presents in the element is :-

(1) 10 (2) 7 (3) 6 (4) 4

57. Any nf-orbital can accomodate upto :-

- (1) 14 electron
(2) Six electrons
(3) Two electrons with parallel spin
(4) Two electrons with opposite spin

58. ${}_{36}\text{Kr}$ has the electronic configuration $({}_{18}\text{Ar}) 4s^2 3d^{10} 4p^6$. The 39th electron will go into which one of the following sub-levels :-

(1) 4f (2) 4d (3) 3p (4) 5s

59. The atomic number of an element is 17, the number of orbitals containing electron pairs in the valency shell is:-

(1) 8 (2) 2 (3) 3 (4) 6

60. Electronic configuration $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{}$ has violated :-

- (1) Hund's rule
(2) Pauli's principle
(3) Aufbau principle
(4) $(n + \ell)$ rule

61. A transition metal 'X' has a configuration $[\text{Ar}] 3d^5$ in its + 3 oxidation state. Its atomic number is:-

(1) 22 (2) 26
(3) 28 (4) 19

62. The maximum probability of finding an electron in the d_{xy} orbital is :-

- (1) Along the x-axis
(2) Along the y-axis
(3) At an angle of 45° from the x and y axis
(4) At an angle of 90° from the x and y axis

63. Radius of H-atom in its ground state is 5.3×10^{-11} m. After collision with an electron it is found to have a radius of 21.2×10^{-11} m. What is the principal quantum no. 'n' of the final state of the atom:-

(1) $n = 2$ (2) $n = 3$
(3) $n = 4$ (4) $n = 16$

64. Bohr's theory is not applicable to -

(1) He (2) Li^{2+}
(3) He^{2+} (4) Both (1) & (3)

65. Which is the correct relationship -

- (1) E_1 of H = $1/2 E_2$ of He^+ = $1/3 E_3$ of Li^{2+} = $1/4 E_4$ of Be^{3+}
(2) $E_1(\text{H}) = E_2(\text{He}^+) = E_3(\text{Li}^{2+}) = E_4(\text{Be}^{3+})$
(3) $E_1(\text{H}) = 2E_2(\text{He}^+) = 3E_3(\text{Li}^{2+}) = 4E_4(\text{Be}^{3+})$
(4) No relation

66. If velocity of an electron in 1st orbit of H atom is V, what will be the velocity of 3rd orbit of Li^{+2} -

(1) V (2) $V/3$
(3) $3V$ (4) $9V$

SPACE FOR ROUGH WORK

67. In a certain electronic transition in the hydrogen atoms from an initial state (1) to a final state (2), the difference in the orbit radius ($r_1 - r_2$) is 24 times the first Bohr radius. Identify the transition -

- (1) $5 \rightarrow 1$ (2) $25 \rightarrow 1$
(3) $8 \rightarrow 3$ (4) $7 \rightarrow 5$

68. Match the following -

- (A) Energy of ground state of He^+ (i) -6.04 eV
(B) Potential energy of I orbit of H-atom (ii) -27.2 eV
(C) Kinetic energy of II excited state of He^+ (iii) $8.7 \times 10^{-18} \text{ J}$
(D) Ionisation potential of He^+ (iv) -54.4 eV

- (1) A - (i), B - (ii), C - (iii), D - (iv)
(2) A - (iv), B - (iii), C - (ii), D - (i)
(3) A - (iv), B - (ii), C - (i), D - (iii)
(4) A - (ii), B - (iii), C - (i), D - (iv)

69. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon :-

- (1) 3s (2) 2p
(3) 2s (4) 1s

70. The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen :-

- (1) $5 \rightarrow 3$ (2) $5 \rightarrow 2$
(3) $4 \rightarrow 3$ (4) $4 \rightarrow 2$

71. If λ_0 is the threshold wavelength for photoelectric emission, λ wavelength of light falling on the surface of metal, and m mass of electron, then de Broglie wavelength of emitted electron is :-

(1) $\left[\frac{h(\lambda\lambda_0)}{2mc(\lambda_0 - \lambda)} \right]^{\frac{1}{2}}$ (2) $\left[\frac{h(\lambda_0 - \lambda)}{2mc\lambda\lambda_0} \right]^{\frac{1}{2}}$

(3) $\left[\frac{h(\lambda - \lambda_0)}{2mc\lambda\lambda_0} \right]^{\frac{1}{2}}$ (4) $\left[\frac{h\lambda\lambda_0}{2mc} \right]^{\frac{1}{2}}$

72. Give the correct order of initials T (true) or F (false) for following statements :-

- (I) If an ion has 2 electrons in K shell, 8 electrons in L shell and 6 electrons in M shell, then number of S electrons present in that element is 6.
(II) The maximum number of electrons in a subshell is given by $2n^2$
(III) If electron has magnetic quantum number -1 , then it cannot be present in s-orbital.
(IV) Only one radial node is present in 3p orbital.

- (1) TTFF (2) FFTF
(3) TFFT (4) FFFT

73. The shortest wavelength of He^+ in Balmer series is x , then longest wavelength in the Paschene series of Li^{+2} is :-

- (1) $\frac{36x}{5}$ (2) $\frac{16x}{7}$
(3) $\frac{9x}{5}$ (4) $\frac{5x}{9}$

SPACE FOR ROUGH WORK

74. Given ΔH for the process $\text{Li (g)} \longrightarrow \text{Li}^{+3}(\text{g}) + 3\text{e}^-$ is 19800 kJ/mole & IE_1 for Li is 520 then IE_2 & IE_3 of Li^+ are respectively (approx value) :-

- (1) 11775, 7505 (2) 19280, 520
(3) 11775, 19280 (4) Data insufficient

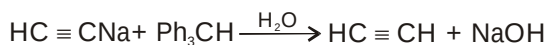
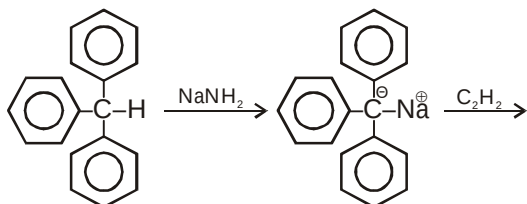
75. Carbanion is a :-

- (1) Base (2) Nucleophile
(3) Both (1) and (2) (4) None

76. Which of the following species has a pyramidal shape-

- (1) CH_3^+ (2) BF_3 (3) CH_3^- (4) $\dot{\text{C}}\text{H}_3$

77. The decreasing order of acid strength indicated by the following sequence of reaction is :-



- (1) $\text{NH}_3 > \text{Ph}_3\text{CH} > \text{C}_2\text{H}_2 > \text{H}_2\text{O}$
(2) $\text{H}_2\text{O} > \text{HC} \equiv \text{CH} > \text{Ph}_3\text{CH} > \text{NH}_3$
(3) $\text{HC} \equiv \text{CH} > \text{H}_2\text{O} > \text{Ph}_3\text{CH} > \text{NH}_3$
(4) $\text{Ph}_3\text{CH} > \text{HC} \equiv \text{CH} > \text{H}_2\text{O} > \text{NH}_3$

78. In which of the following compounds is hydroxylic proton the most acidic -

- (1) (2)
(3) (4)

79. Correct arrangement of the following acids in correct K_a order is :-

- (I) $\text{H}_3\text{N}^+ - \text{CH}_2 - \text{COOH}$
(II) $\text{NC} - \text{CH}_2 - \text{COOH}$
(III) $\text{H}_3\text{C} - \text{CH}_2 - \text{COOH}$
(IV) $^-\text{OOC} - \text{CH}_2 - \text{COOH}$

- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$
(2) $\text{II} > \text{I} > \text{III} > \text{IV}$
(3) $\text{I} > \text{III} > \text{II} > \text{IV}$
(4) $\text{IV} > \text{III} > \text{II} > \text{I}$

80. Which of the following orders of acid strength is correct :-

- (1) $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC} \equiv \text{CH}$
(2) $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC} \equiv \text{CH}$
(3) $\text{RCOOH} > \text{HOH} > \text{HC} \equiv \text{CH} > \text{ROH}$
(4) $\text{RCOOH} > \text{HC} \equiv \text{CH} > \text{HOH} > \text{ROH}$

81. Which of them is false for order of - I effect

- (1) $-\text{F} > -\text{Cl} > -\text{Br} > \text{I}$
(2) $-\text{NR}_3^+ > -\text{NH}_3^+ > -\text{NO}_2$
(3) $-\text{OCH}_3 > -\text{OH} > -\text{NH}_2$

- (4) $> -\text{C} \equiv \text{CH} > \text{H}$

82. The order of basicity of halides is :

- (1) $\text{Cl}^- < \text{Br}^- < \text{I}^- < \text{F}^-$
(2) $\text{F}^- < \text{I}^- < \text{Br}^- < \text{Cl}^-$
(3) $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$
(4) $\text{Cl}^- < \text{F}^- < \text{I}^- < \text{Br}^-$

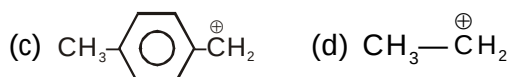
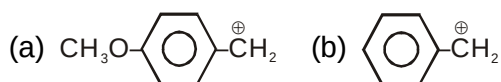
SPACE FOR ROUGH WORK

83. Which of the following molecule has longest C=C bond length -

- (1) $\text{CH}_2=\text{C}=\text{CH}_2$ (2) $\text{CH}_3-\text{CH}=\text{CH}_2$

- (3) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}=\text{CH}_2$ (4) $\text{CH}_3-\overset{\text{CH}_3}{\text{C}}=\text{CH}_2$

84. Consider the following carbocations



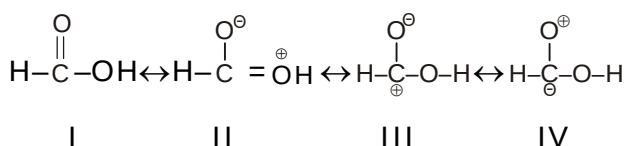
The relative stabilities of these carbocations are such that :-

- (1) $d < b < c < a$ (2) $b < d < c < a$
(3) $d < b < a < c$ (4) $b < d < a < c$

85. Among the following, the strongest base is :-

- (1) $\text{C}_6\text{H}_5\text{NH}_2$
(2) $p\text{-NO}_2-\text{C}_6\text{H}_4\text{NH}_2$
(3) $m\text{-NO}_2-\text{C}_6\text{H}_4\text{NH}_2$
(4) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

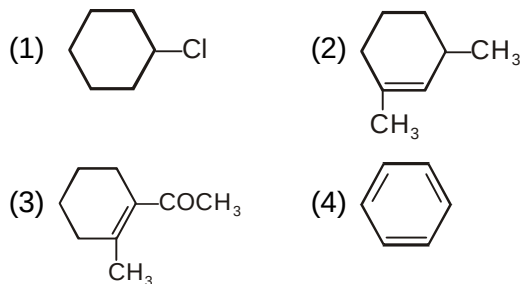
86. Formic acid is considered as a hybrid of the four structures



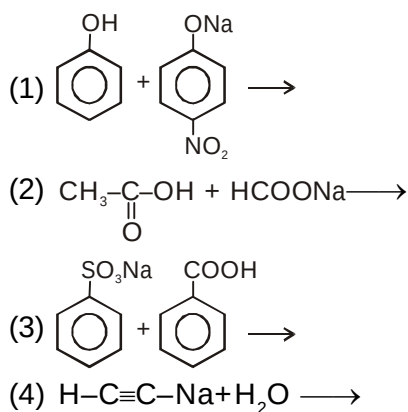
Which of the following order is correct for the stability of the four contributing structures.

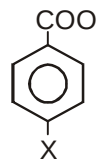
- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{I} > \text{II} > \text{IV} > \text{III}$
(3) $\text{I} > \text{III} > \text{II} > \text{IV}$ (4) $\text{I} > \text{IV} > \text{III} > \text{II}$

87. In which of the following molecule all the effect namely inductive, mesomeric & hyperconjugation operate:



88. Which of the following reaction is possible :



89.  , pK_a value of the compound decreases if X is :-

- (1) $-\text{NO}_2$ (2) $-\text{NH}_2$
(3) $-\text{OH}$ (4) $-\text{OCH}_3$

90. Energy of activation is lower for which reaction :

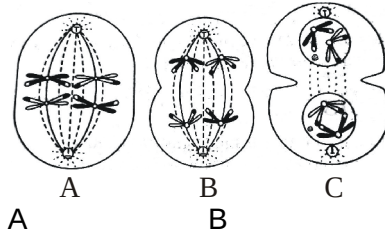
- (1) $\text{RCH}_2\text{OH}_2^+ \rightarrow \text{RCH}_2^+$
 (2) $\text{R}_2\text{CHOH}_2^+ \rightarrow \text{R}_2\text{CH}^+$
 (3) $\text{R}_3\text{COH}_2^+ \rightarrow \text{R}_3\text{C}^+$
 (4) all have same E_{act} .

SPACE FOR ROUGH WORK

91. Nucleus is absent in:
 (1) Mature RBC
 (2) mature sieve tube
 (3) Both (1) and (2)
 (4) All mature cells
92. Sometimes a few chromosomes have non-staining secondary constriction at a constant location. This gives the appearance of a small fragment beyond the secondary constriction called:
 (1) Centriole
 (2) Centromere
 (3) Satellite
 (4) Both (2) and (3)
93. Which one is correct for heterochromatin of eukaryotic nucleus?
 (1) It stains densely with basic dyes
 (2) It is highly condensed in interphase
 (3) It is highly expanded in interphase, stains densely with acidic dyes
 (4) Both (1) and (2)
94. Which of the following is correct for G_1 -phase
 (1) Corresponds to the interval between mitosis and initiation of DNA replication
 (2) Histone protein is synthesised
 (3) Centriolar duplication occurs
 (4) All of these
95. G_0 or quiescent stage is:
 (1) Permanent exit of cell from cell cycle
 (2) Temporary exit of cell from cell cycle
 (3) A metabolically inactive stage
 (4) Either temporary or permanent exit of cell from cell cycle
96. I. Cells like heart cells and nerve cells, which are also called non-dividing cells are inA.... phase.
 II. During S-phase in animal cells, DNA replication takes place in theB..... and duplication of centriole takes place in theC.... .

Fill in the blanks with appropriate option.

- (1) A- G_0 , B-nucleus, C-cytoplasm
 (2) A-S, B-cytoplasm, C-nucleus
 (3) A- G_1 , B-nucleus, C-cytoplasm
 (4) A- G_0 , B-cytoplasm, C-nucleus
97. Identify the given figures showing meiotic phases and select the correct option.



- A B C
 (1) Metaphase Anaphase Telophase
 (2) Metaphase-I Anaphase-I Telophase-I
 (3) Metaphase-II Anaphase-II Telophase-II
 (4) Anaphase-I Metaphase-I Telophase-I

98. Match the following columns.

Column I	Column II
A. Terminalisation	1. Metaphase-I
B. Histone synthesis	2. Anaphase-I
C. Disjunction	3. Diakinesis
D. Interkinesis	4. Meiosis-II
E. Double equatorial plate	5. S-phase

Codes

A	B	C	D	E
(1) 5	2	1	3	4
(2) 3	5	4	2	1
(3) 2	5	4	1	3
(4) 3	5	2	4	1

99. Crossing over occurs during :-
 (1) Diplotene (2) Pachytene
 (3) Zygotene (4) Leptotene
100. In Meiosis the homologous chromosomes are separated during :-
 (1) Metaphase I (2) Prophase II
 (3) Anaphase I (4) Anaphase II
101. Chromosome number is reduced to half in :-
 (1) Equational division
 (2) Meiosis II
 (3) Homotypic division
 (4) Meiosis I

102. In plant cytokinesis, cell plate grows :-

- (1) Centripetally
- (2) Centrifugally
- (3) Atrandom
- (4) Terminally

103. DNA replication in bacteria occurs

- (1) Within nucleolus
- (2) Prior to fission
- (3) Just before transcription
- (4) During S phase

104. Mitochondria DNA differs from the nuclear DNA in :-

- (1) Lacking association with histone proteins
- (2) Being circular in nature
- (3) Having higher C-G ratio
- (4) All of these

105. What happens during growth of a cell :-

- (1) K.I. decreases
- (2) K.I. increases
- (3) K.I. fluctuates
- (4) K.I. remains constant

106. What is the relationship among DNA, a gene, and a chromosome?

- (1) A chromosome contains hundreds of genes, which are composed of DNA.
- (2) A chromosome contains hundreds of gene, which are composed of protein.
- (3) A gene contains hundreds of chromosomes, which are composed of protein.
- (4) A gene contains hundreds of chromosomes, which are composed of DNA.

107. The significance of mitosis is that:

- (1) It helps in repair
- (2) It replaces the cells of lining of the gut and blood cells
- (3) It helps in continuous growth of plants at meristematic points
- (4) More than one option is correct

108. Which of the following lack both mitosis and meiosis?

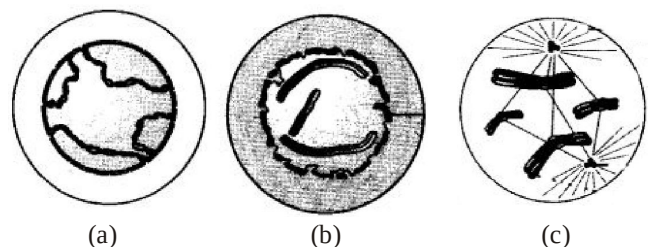
- (1) Eubacteria
- (2) Blue green algae
- (3) Mycoplasmas
- (4) More then one option is correct

109. Which is not a characteristic of telophase?

1. Chromatin condenses to form chromosome
2. Nucleolus, Golgi complex and ER reform
3. Nuclear envelope assemble around chromosome clusters
4. Centromeres split and chromatids separate
5. Chromosome clusters and their identity is lost

- (1) 1, 2, 4 only
- (2) 1, 4 only
- (3) 2, 3 only
- (4) 3, 4, 5 only

110. Recognise the figure and find out the correct matching :-

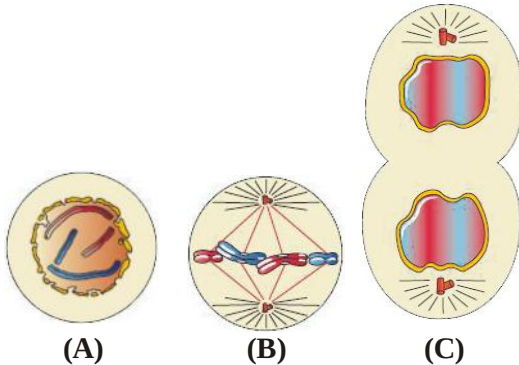


- (1) a–Early prophase, b–late prophase, c–transition to metaphase
- (2) b–Early prophase, c–late prophase, a–transition to metaphase
- (3) c–Early prophase, a–late prophase, b–transition to metaphase
- (4) b–Early prophase, a–late prophase, c–transition to metaphase

111. A bivalent consists of

- (1) Two chromatids and one centromere
- (2) Four chromatids and two centromeres
- (3) Two chromatids and two centromeres
- (4) Four chromatids and four centromeres

112. The figures below show 3 phases of mitosis select the option giving correct identification together with the correct event ?

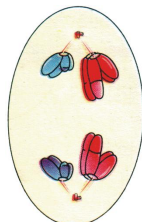


- (A) C-Telophase-nuclear envelope assembles around the chromosome clusters
(2) B-Anaphase-segregation of homologous chromosome
(3) A-Prophase-chromosomes get fully condensed
(4) C-Metaphase-condensation of chromatin to form chromosome

113. At which stage of meiosis, do these events occurs ?

	Bivalents more clearly visible	Formation of synaptonemal complex	Terminalisation of chiasmata
(1)	Leptotene	Zygotene	Pachytene
(2)	Zygotene	Pachytene	Diakinesis
(3)	Pachytene	Zygotene	Diakinesis
(4)	Zygotene	Diplotene	Diakinesis

114. Identify the below stage with their respective event :-



- (1) Anaphase-II Splitting of the centromere
(2) Anaphase-I Homologous chromosome separate
(3) Anaphase-I Splitting of the centromere
(4) Metaphase-I Bivalent chromosomes align on the equatorial plate.

115. In which of the following mitosis does not occur?

- (1) Fungi (2) Bacteria
(3) Green algae (4) Higher plants

116. During the G_1 phase of cell division:

- (1) Cell prepares for M phase.
(2) Cell undergoes duplication.
(3) RNA and proteins are synthesized.
(4) DNA and proteins are synthesized.

117. Which type of DNA is present in the G_2 phase of cell cycle?

- (1) 1 stranded (2) 2 stranded
(3) 4 stranded (4) 8 stranded

118. It is difficult to observe individual chromosomes during interphase because:

- (1) the DNA has not been replicated yet.
(2) they have uncoiled to form long, thin strands.
(3) homologous chromosomes do not pair up until division starts.
(4) the spindle must move them to the metaphase plate before they become visible.

119. A very significant contribution of mitosis is cell repair. Repairing takes place in our body in

- (1) Bone cells
(2) Upper layers of epidermis
(3) Cells of the lining of the gut
(4) All of the above

120. Match column-I with column-II and select the correct option from the codes given below.

Column A		Column B	
A.	NOR	I	Secondary constriction
B.	Nuclear lamina	II	MTOC
C.	Histone	III	Intermediate filament
D.	Centrosome	IV	Nucleosome

Options :-

- (1) A-I, B-III, C-IV, D-II
(2) A-I, B-III, C-II, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-IV, C-II, D-III

121. Interphase includes all except
 (1) G_1 phase (2) Anaphase
 (3) S phase (4) G_2 phase
122. Chromosome condensation occurs in:-
 (1) G_1 phase (2) S phase
 (3) G_2 phase (4) M phase
123. Identify the stage in which chromosomes replicate while the sister chromatids remain associated at their centromeres.
 (1) G_1 phase (2) S phase
 (3) G_2 phase (4) M phase
124. Every chromosomes essentially has a ____A____ on the sides of which disc shaped structures called ____B____ are present.
 (1) A - Secondary constriction
 B - Kinetochore
 (2) A - Primary constriction
 B - Kinetochore
 (3) A - Kinetochore
 B - Secondary constriction
 (4) A - Kinetochore
 B - Primary constriction
125. The active phase, also called metabolic or energetic phase with most cytogenetic activity is
 (1) Interphase (2) M-Phase
 (3) Meiosis (4) Pachytene
126. Bouquet stage shows convergence of chromosomal ends towards centriole during
 (1) zygotene (2) leptotene
 (3) pachytene (4) diplotene
127. The best stage to view structure, size and to count the number of chromosomes is
 (1) Metaphase (2) Late prophase
 (3) Telophase (4) I-phase
128. Synaptonemal complex is a ribonucleo-protein structure reported by Moses, 1956. It is visible from
 (1) leptotene through diplotene
 (2) pachytene through diplotene
 (3) zygotene through pachytene
 (4) diplotene through metaphase
129. What is the cause of cell division?
 (1) Increase in surface area/volume ratio due to growth of cell.
 (2) Nucleo-cytoplasmic ratio decreases due to increase in size of cell.
 (3) Disturbance in lipid and protein ratio
 (4) All of the above
130. The sequence of substages of prophase I are
 (1) Leptotene — Zygotene — Pachytene — Diplotene — Diakinesis
 (2) Leptotene — Zygotene — Pachytene — Diakinesis — Diplotene
 (3) Leptotene — Pachytene — Zygotene — Diplotene — Diakinesis
 (4) Leptotene — Pachytene — Diplotene — Zygotene — Diakinesis
131. Actual haploidy in terms of DNA comes in
 (1) metaphase I
 (2) anaphase
 (3) anaphase II
 (4) cell plate
132. The mitosis and meiosis are unlike to each other because in meiosis
 (1) chromosomes first pair and then reduced to half
 (2) homologous chromosomes pair and exchange parts
 (3) 4 nuclei are formed which are not identical
 (4) all of the above three are correct
133. The enzyme recombinase is required at which stage of meiosis
 (1) Diakinesis
 (2) Pachytene
 (3) Zygotene
 (4) Diplotene

134. Which of the following options gives the correct sequence of events during mitosis?

- (1) Condensation → nuclear membrane disassembly → crossing over → Segregation → telophase
- (2) Condensation → nuclear membrane disassembly → arrangement at equator → centromere division → segregation → telophase
- (3) Condensation → crossing over → nuclear membrane disassembly → segregation → telophase
- (4) Condensation → arrangement at equator → centromere division → segregation → telophase

135. Nuclear envelope is a derivative of

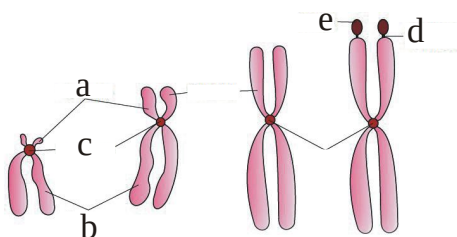
- (1) membrane of Golgi complex
- (2) microtubules
- (3) rough endoplasmic reticulum
- (4) smooth endoplasmic reticulum

136. Match the columns and identify the correct option

Column-I	Column-II
1 Amitosis	(i) Prokaryotes
2 Closed mitosis	(ii) Lower Eukaryotes
3 Mitosis	(iii) Diploid cell
4 Meiosis	(iv) Haploid cell

- (1) (2) (3) (4)
- (1) (iv) (iii) (i) (ii)
- (2) (i) (ii) (iii) (iv)
- (3) (i) (ii) (iv) (iii)
- (4) (iii) (iv) (ii) (i)

137. Recognize the figure and find the correct match



- (1) a-p arm, b-long arm, c-primary constriction, d-SAT
- (2) a-short arm, b-q arm, c-centromere, d-secondary constriction
- (3) a-q arm, b-p arm, c-centromere, d-satellite
- (4) a-p arm, b-q arm, c-primary constriction, e-secondary constriction

138. What % of trihybrid will be obtained by the cross of trihybrid plant with a recessive plant :-

- (1) 25% (2) 12.5%
- (3) $\frac{1}{64} \times 100$ (4) $\frac{3}{8} \times 100$

139. The genes for male sterility & albinism in plants, are found respectively in :-

- (1) Mitochondria & Plastid
- (2) Plastid & Mitochondria
- (3) Plastid & Plastid
- (4) None of the above

140. Skin colour in human, is controlled by three gene pairs, then what is the number of intermediate & parental combinations in F_2 generation respectively :-

- (1) 2, 62 (2) 62, 2
- (3) 2, 14 (4) 14, 2

141. Which of the following type of inheritance can we find out with the help of reciprocal cross?

- (1) Cytoplasmic inheritance
- (2) Gene inheritance
- (3) Maternal inheritance
- (4) (1) & (3) both

142. Genes a, b and c assort independently and are recessive to their respective alleles A, B and C. Two triple heterozygous AaBbCc individual are crossed what is the probability that a given offspring will be genotypically homozygous for all three dominant alleles?

- (1) 2/64 (2) 27/64
- (3) 1/64 (4) 6/64

143. A dihybrid plant is with genotype GgTt. In this plant first gene pair is codominant on selfing of this plant what is phenotypic ratio?
- (1) 2 : 1 (2) 3 : 1
(3) 3 : 6 : 3 : 1 : 2 : 1 (4) 6 : 3 : 2 : 1
144. Which will not the genotype of pollen grain of a plant with genotype HHbbDdnnMmaa?
- (1) HbdNMa (2) HbDnMa
(3) HbdnMa (4) All the above
145. The allele, which is unable to express its effect in the presence of another is called :-
- (1) Co-dominant (2) Supplementary
(3) Complementary (4) Recessive
146. In the inheritance of flower colour in dog flower plant, the F_1 had a phenotype that
- (1) Resembles both of the parents
(2) Did not resemble either of the two parents
(3) Resemble with only one parent
(4) (1) & (3) both
147. The three different alleles of human ABO blood types will produce how many genotypes & phenotypes respectively :-
- (1) 4 & 6 (2) 6 & 4
(3) 6 & 6 (4) 4 & 4
148. Other than pea plants it was found that sometimes the F_1 had a phenotype that did not resemble either of the two parents and was in between the two. Due to :-
- (1) Complete dominance
(2) Incomplete dominance
(3) Co-dominance
(4) Complementary gene interaction
149. Which of the following condition is true for co-dominance :-
- (1) Phenotype of F_1 resembled either of the two parents
(2) Phenotype of F_1 did not resemble either of two parents
(3) Phenotype of F_1 resembles both parents
(4) None of these

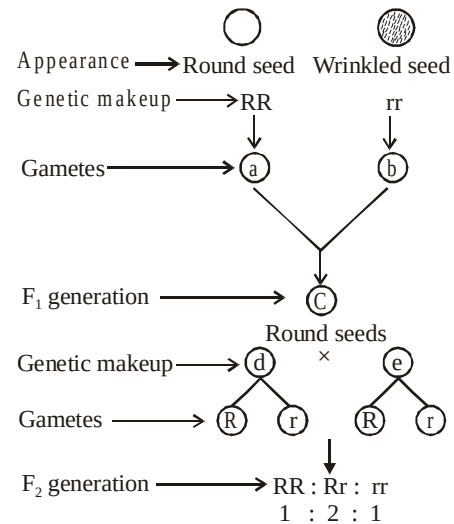
150. Which of the following is a good example of multiple allele :-
- (1) ABO blood group
(2) Size of starch grain in pea
(3) Shape of seed
(4) Flower colour in pea
151. In mice, black coat colour (allele B) is dominant to brown coat colour (allele b). The offspring of a cross between a black mouse (BB) and a brown mouse (bb) were allowed to interbreed. What percentage of the progeny would have black coats ?
- (1) 25 % (2) 50 %
(3) 75 % (4) 100 %
152. Fruit shape in shepherd's purse (Capsella bursa) is of two types—triangular and top-shaped. Triangular fruit shape (T) is dominant over top-shape (t). Following table summarizes the results of several crosses.

Cross	Result
Strain 1 $\times$ tt	All triangular
Strain 2 $\times$ tt	1 triangular : 1 top – shaped
Strain 3 $\times$ tt	All top – shaped
Strain 4 $\times$ Tt	3 triangular : 1 top – shaped

Which pair of strains possess the genotype Tt?

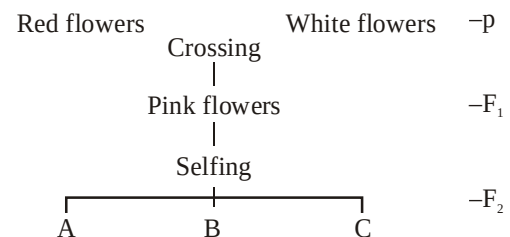
- (1) Strains 2 and 3
(2) Strains 2 and 4
(3) Strains 1 and 3
(4) Strains 1 and 4
153. Which one of the following crosses would have 1 : 1 : 1 : 1 ratio ?
- (1) TtRR $\times$ ttrr
(2) TTRR $\times$ ttrr
(3) TtRr $\times$ ttrr
(4) TtRR $\times$ TTrr
154. Possible blood group in children from the parents with 'B' and 'O' blood group are :
- (1) All B
(2) All O
(3) Both B and O
(4) A and B

155. Reciprocal cross is
- (1) Intraspecific hybridization
 - (2) Back cross
 - (3) Pollen grains from one variety deposited on the stigma of the contrasting variety and vice-versa, in hybridization experiments
 - (4) Test cross
156. Complementary genes were demonstrated by Bateson and Punnett in
- (1) *Capsella*
 - (2) *Lathyrus odoratus*
 - (3) *Antirrhinum*
 - (4) *Mirabilis*
157. Which of the following genotype of sweet pea plant is related with the production of purple coloured flowers ?
- (1) CcPp
 - (2) CCpp
 - (3) ccPP
 - (4) Ccpp
158. Total number of genotypes and phenotypes in a dihybrid test cross is -
- (1) 9
 - (2) 8
 - (3) 16
 - (4) 13
159. Read the following statements and identify correct ones :-
- (I) Gene that control seed shape in garden pea is a pleiotropic gene.
 - (II) Starch grain size in garden pea seeds is a codominant character.
 - (III) Seed shape in garden pea is regulated by SBE gene.
- (1) I, II and III
 - (2) I and III
 - (3) Only II
 - (4) II and III
160. In pea plant, genotype (Bb) shows -
- (1) Round seed with large sized starch grain
 - (2) Round seed with intermediate sized starch grain
 - (3) Round seed with small sized starch grain
 - (4) Wrinkled seed with small sized starch
161. Given below is the diagram of a monohybrid cross between round and wrinkled seeded pea plants. The round allele (R) is dominant over wrinkled allele (r). From the given diagram identify a, b, c, d & e :-



- (1) a = R, b = r, c = RR, d = rr, e = RR
- (2) a = R, b = r, c = Rr, d = rr, e = RR
- (3) a = r, b = R, c = rr, d = Rr, e = RR
- (4) a = R, b = r, c = Rr, d = Rr, e = Rr

162. A cross is being given here



If B stands for pink flower then

- (a) What is the name of plant ?
 - (b) What is percentage of A in F₂ progeny ?
- (1) (a) Garden pea, (b) 25%
 - (2) (a) *Mirabilis*, (b) 25%
 - (3) (a) Garden pea, (b) 50%
 - (4) (a) *Snapdragon*, (b) 50%
163. Two plants of garden pea were crossed and 600 plants were obtained in progeny. Out of these 310 plants produced round-seeds. While rest produced wrinkle seeds. What is the genotype of crossed plants ?
- (1) Rr × rr
 - (2) RR × rr
 - (3) Rr × Rr
 - (4) Rr × RR
164. Blood group of father is B and that of daughter is AB. The genotype of mother could be :-
- (1) I^AI^A or ii
 - (2) I^AI^B or ii
 - (3) I^Ai or I^AI^A or I^AI^B
 - (4) I^AI^A or ii or I^AI^B

165. In a dihybrid cross 640 plants were produced in F_2 . If first character shows dominance and the second character shows incomplete dominance then how many of these plants are showing dominant trait for first characters and intermediate trait for second character?
(1) 240 (2) 120 (3) 310 (4) 160
166. Let genes A & B both be necessary for normal hearing. What is the possible genotype of normal child of a deaf mother & a deaf father ?
(1) Aabb (2) aaBB
(3) aabb (4) AaBb
167. Which of the following is incorrect match ?
(1) T.H. Morgan – discovery of linkage
(2) A Garrod – study of Alkaptonuria
(3) W. Bateson – father of experimental genetics
(4) Carl Correns–discovery of incomplete dominance
168. If a character is controlled by four alleles of a gene, then the possible genotypes would be :-
(1) 10 (2) 15 (3) 20 (4) 21
169. $\frac{1}{4} : \frac{1}{2} : \frac{1}{4}$ ratio of TT : Tt : tt can be depicted mathematically binomial expression as (ideally) :-
(1) $(ax + by)^2$ (2) $(ax + by)^3$
(3) $(Ax + By)^4$ (4) $ax + by$
170. In dominant epistasis, dominant allele is epistatic over hypostatic gene. In summer squash (*Cucurbita pepo*) gene for white fruit colour (W) is epistatic over yellow (Y) which is dominant over green (y). What will be the ratio of fruit colour in this given cross ? $WwYy(\text{white}) \times Wwyy(\text{White})$
(1) 6 white : 1 yellow : 1 green
(2) 12 white : 3 yellow : 3 green
(3) 12 white : 3 yellow : 1 green
(4) 9 white : 3 yellow : 4 green
171. In a plant weight of fruit is controlled by three genes, the maximum and minimum weights possible are 66 gm and 30 gm. What must the weight of a fruit produced from a plant having AaBBCc genotype ?
(1) 56 gm (2) 54 gm
(3) 46 gm (4) 44 gm
172. In Sickle cell anaemia mutation :
(1) Valine is replaced by glutamic acid
(2) Heterozygous genotype is lethal
(3) Homozygous genotype is lethal
(4) RBC become nucleated
173. In case of human skin colour inheritance ratio:
 $1 : \boxed{A} : \boxed{B} : 20 : \boxed{B} : \boxed{A} : 1$
A and B are :
(1) A = 4, B = 15
(2) A = 6, B = 12
(3) A = 4, B = 12
(4) A = 6, B = 15
174. In case of mice, 'Y' controls yellow body colour and it also shows lethality. Which genotype must result in lethality ?
(1) YY and Yy (2) YY only
(3) Yy and yy (4) yy only
175. Hair colour in dog shows dominant epistasis :
B = black hair colour
b = brown hair colour
I = Epistatic gene – White hair colour
Crossing ♂ and ♀ dog of genotype BbIi must give :
(1) 12-black, 3-white, 1-brown
(2) 12-white, 3-black, 1-brown
(3) 12-brown, 3-white, 1-black
(4) 12-white, 3-brown, 1-black
176. A man of A blood group marries a woman of AB blood group. Which type of progeny would indicate that man heterozygous ?
(1) O (2) B
(3) AB (4) A

177. Mice having yellow body colour is crossed with brown colour one. What percentage of progeny show lethality ?

- (1) 25% (2) 75%
(3) 0% (4) 100%

178. In given diagram which of the following combination showing recessive epistasis and supplementary influence :-

F₂

			
A	B	C	D
			
E	F	G	H
			
I	J	K	L
			
M	N	O	P

Agouti : Black : Albino
9 : 3 : 4

	Recessive epistasis	Supplementary
(1)	K L O	F H N P
(2)	F N H P	K L O
(3)	F N H P	A B C D
(4)	K L O	A B C D

179. In a population, polygenic traits give?

- (1) Linear curve
(2) Hyperbolic curve
(3) Bell shaped curve
(4) Sigmoid curve

180. If dominant C and P genes are essential for the development of purple colour in *Lathyrus odoratus* flowers. What would be the ratio of purple and white colour in a cross between CcPp × cc Pp.

- (1) 3 : 5
(2) 9 : 7
(3) 2 : 6
(4) 4 : 4

SPACE FOR ROUGH WORK



Result : NEET (UG) 2019

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